there should be no content-specific activation. This translates in expected temporal generalization of decoding accuracy between the onset and offset burst with no generalization in between. This prediction can be translated in a model matrix to be correlated with the obtained temporal generalization results. IIT predicts sustained, content-specific activity patterns in posterior cortex throughout the duration of the percept (after the initial transient response). Therefore, activation levels within posterior areas should be higher for longer-duration (and consciously seen) than shorter-duration (and no longer seen) stimuli during analysis time windows 2 and 3. In addition, temporal generalization of decoding accuracy should be evident for as long as the stimuli are consciously experienced, and will be tested via correlation with a different model matrix than for GNW.

3. Inter-areal communication: Patterns of synchronization. This analysis will focus on information sharing/integration between different cortical regions for each specific stimulus category.

GNW predicts that conscious perception, regardless of task, relies on long-range information sharing between nodes of PFC and Fusiform Face Area (FFA) when seeing a face, and PFC and Lateral Occipital Cortex (LOC) when seeing an object. This implies that synchronization within these regions should be more consistent across tasks (for a given stimulus category) than across stimuli (for a given task). IIT predicts strong bindings ('relations') between category-selective areas and lower-level visual areas, which again should be found regardless of the task. Hence, patterns of synchronization should be found between category-selective regions (FFA and LOC) and early visual cortex: specifically, between FFA and V1/V2 when seen faces (but not other stimulus categories) and between LOC and V1/V2 when seen objects (but not other stimulus categories). This synchronization should be more consistent across tasks than across stimuli.

4. Putative NCC analysis. We will also perform further analyses to delineate the putative NCC for which GNW and IIT provide different predictions (PFC vs. posterior hot zone, respectively), after ruling out areas based on task contrasts (see the OSF preregistration for more details). To that end, we will run three contrast/conjunction analyses, which can be performed on univariate fMRI activation maps as well as on multivariate fMRI decoding maps: (A) Areas that are sensitive to *task goal (attending*, *detecting*, *& responding to target stimuli)*, showing greater activity for task-relevant targets vs. baseline (blank inter-trial intervals), and–importantly–no differential activity for non-targets (both for task-relevant or task-irrelevant) vs. baseline (blank ITIs); (B) Areas that are sensitive to *task-relevance (attending & detecting stimuli of the relevant categories)*, and accordingly are responsive to all task-relevant stimuli, whether targets or not, but are not responsive to task-irrelevant stimuli; (C) *candidate areas for enabling conscious perception (putative NCCs)*. We will look for the conjunction of areas sensitive to changes in the content of consciousness (stimulus present vs. blank ITIs) within each stimulus category (e.g., faces vs. blank, objects vs. blank, letters vs. blank, false fonts vs. blank).

The first two analyses (A & B) are aimed at *identifying areas that are most likely involved in the consequences of consciousness and are unlikely to be related to neural processes mediating consciousness per se*. The third analysis (C) is aimed at identifying *areas that may contain NCCs and/or sensory precursors to visual consciousness*. We note that the proposed contrasts might underestimate the consequences of consciousness (analysis A and B) and overestimate the NCC (analysis C). We have adopted a conservative approach which has the advantage of firmly, while not exhaustively, distinguishing between areas that might participate in consciousness vs. those that definitely do not. GNW predicts that the putative NCC should include prefrontal cortex after ruling out areas based on task contrasts. IIT in turn predicts that the putative NCC should be mapped primarily onto posterior areas, after ruling out areas based on task contrasts.

Pilot data. To establish the adequacy of our experimental protocols, this experiment has been pilot-tested for behavioral and eye movement patterns. The results of these pilot studies are reported on the OSF page (https://osf.io/6vgy3/). We also conducted a control experiment testing the visibility of the stimuli in an offline, surprise memory test. These pilot tests showed that subjects detect the targets with high sensitivity and specificity while being able to maintain fixation on the stimulus throughout the duration of the trial. They further confirmed that subjects were indeed aware of all stimuli, including the task-irrelevant ones, as performance and confidence for task-irrelevant stimuli was overall similar to task-relevant stimuli.

Experiment 2

Aim. The general goal of Experiment 2 is to measure brain activity elicited by salient visual stimuli that are reported as seen versus unseen in the context of a secondary task. Stimulus visibility is manipulated via attention, by using an engaging video game as a primary, attentionally-taxing task. As such, this experiment predominantly evaluates the neural mechanisms underlying conscious vs. unconscious processing. Several predictions of GNW and IIT will be tested. GNW's hypotheses are directly related to the contrast between reported-seen and reported-unseen trials, while IIT's predictions focus on seen trials across different task manipulations. The most critical tests of GNW relate to the decodability of conscious content in prefrontal areas for seen stimuli, regardless of task manipulations, as well as the prediction that ignition of the global workspace (information sharing)–as measured by long-range synchrony between the prefrontal and sensory cortices–is key to conscious perception (again, regardless of the task). IIT's central prediction in this experiment is that activity and patterns of connectivity within posterior cortical areas will be consistent across all task conditions for seen stimuli triggering similar experiences (within the same category), but different for seen stimuli triggering different experiences (across different categories), independently of the task in which those seen stimuli are embedded.

Design, stimuli, and procedure. We seek to measure brain activity elicited by salient stimuli reported as seen versus unseen due to inattentional blindness [55], in an attentional-taxing, engaging video game context. To minimize the relevance of the critical stimuli, subjects will be challenged to maximize performance on the video game, with visibility reports serving as a de-emphasized secondary task. This experiment will focus on investigating differences in activation, decoding, and level of inter-regional synchronization between seen and unseen stimuli, including prestimulus responses as well as early vs. late post-stimulus responses.

Large (2.3˚), high contrast, faces and objects will be presented for 250 ms at one of four locations (two in the left, two in the right visual field, 6.4˚ eccentricity), while scrambled texture patterns [56] made from superimposing faces and objects will be presented at the other three locations. The stimuli will be displayed on top of rotating square shapes in the background of a video game, which spans the entire screen (to estimate the relative size of the stimuli compared with the background, see Fig 3).

Identical stimuli will be presented during a *distracted attention* (dAT) and an *attended* (AT) task. In the former, subjects will be instructed to exclusively focus on playing an engaging video game (for a detailed description, see the preregistration and the demo movie: https://osf.io/b9nce/) while infrequent "probes" will assess whether they consciously perceived a subset of the face and object stimuli presented in the background (see Fig 3 for further description of the videogame). The probes will be presented while the game is paused with an arrow cue pointing towards one of the potential stimulus locations. Subjects will be instructed to respond yes/no to each probe by pressing one of two keys to indicate if they saw a stimulus at that location or not. There will be two types of probes. "Awareness probes" will follow a stimulus presentation and

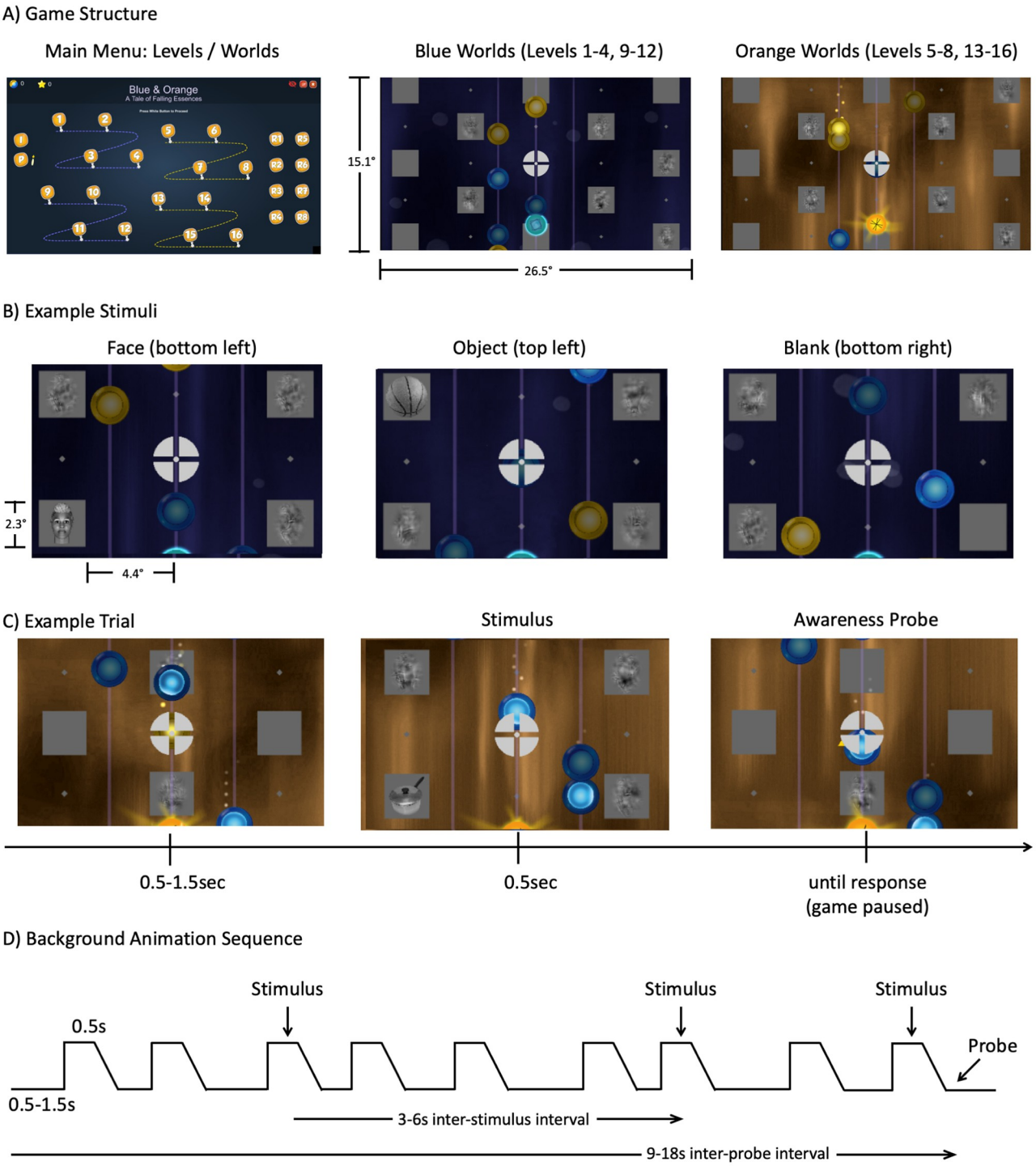

Fig 3. Experiment 2 design–A) The four worlds of the game, with four levels in each world (left). In worlds 1 & 3 (middle) the player will control a shiny blue orb at the bottom of the screen to collect falling disks of the same color (on three vertical tracks), while avoiding falling disks of a different color. The size of the full game display in visual angle is indicated here (panel A, middle). In worlds 2 & 4 the colors will be swapped and players will control an orange orb (right). B) Examples of stimuli presented in the background (central portion of screen blown-up for display purposes): face (left), object (middle), blank (right). C) During gameplay, at 3-6sec intervals (4-7sec for fMRI), an object/face/blank will appear on one of four background squares (middle) for 500ms total, 250ms at full contrast, followed by 250ms of fade-out. On some trials (every 9-18sec), the stimulus will be immediately followed by an awareness probe (right) in which the game will pause and a small arrow will appear in the center instructing subjects to report whether they had just noticed a stimulus in that location. D) Schematic of the background animation demonstrating the trial timing, stimulus timing, and probe timing. For a video demonstration of the video game, visit the preregistration website: https://osf.io/b9nce/).

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